Kestrel M2 — User Manual

Revision 4.2 · Kestrel M2 and Kestrel M2 Pro desktop 3D printers

1. In the box

The Kestrel M2 ships assembled. Lift it out of the foam by the frame, never by the gantry. Inside you will find the printer, a spool holder, a spare 0.4 mm brass nozzle, a 200 g sample spool of grey PLA, a microSD card, the power cable for your region, and a tool roll containing hex keys, a nozzle spanner, a scraper and a nylon cold-pull filament.

Remove all four orange transit clips before you switch the printer on. Two hold the gantry to the uprights and two hold the bed carriage. The printer will report a motion fault if you leave them in place.

2. First switch-on

1. Place the printer on a flat, solid surface with 100 mm of clearance behind it.
2. Fit the spool holder to the left upright and hang the sample spool on it.
3. Connect the power cable and set the switch on the back to I.
4. Follow the on-screen setup: language, then the self test, then the first levelling.

The self test moves each axis to its limit and back and takes about two minutes. Do not touch the printer while it runs.

3. Levelling the bed

The Kestrel M2 probes the plate automatically. You never turn a levelling screw; the firmware builds a height map and compensates as it prints. Run a levelling in three situations: after moving the printer, after changing the build plate, and whenever a first layer looks uneven.

Running a levelling

1. Wipe the plate with isopropyl alcohol and let it dry.
2. Choose *Setup* → *Levelling* → *Full mesh* from the panel.
3. The printer heats the bed to 60 °C, then probes forty-nine points. It takes four minutes.
4. The mesh is stored. You do not need to repeat it before every print.

If probing fails at one point the printer stops with E-19. A single point failing at the front left is almost always a loose probe cable.

4. The first layer

Almost every print problem people bring to us is a first-layer problem, and almost every first-layer problem is one of three things: a dirty plate, a Z-offset that is wrong by a few hundredths of a millimetre, or a bed that is too cold for the material.

If the first layer is not sticking

1. Wash the plate with warm water and dish soap, and dry it with a clean cloth. Skin oil from handling the plate is the most common cause of a first layer lifting, and isopropyl alcohol alone does not remove it.
2. Check the bed temperature against the materials guide. 60 °C for PLA, 80 °C for PETG.
3. Lower the Z-offset by 0.05 mm and print the test square again. A first layer that sticks looks slightly squashed, with no gaps between the lines.
4. Turn the part cooling fan off for the first two layers.
5. Print onto a raft or a brim if the part has a small footprint.

Adjust the Z-offset live from *Tune → Z-offset* while the first layer is printing. The change is saved when the print finishes.

If the first layer is too flat or ridged

The nozzle is too close. Raise the Z-offset by 0.05 mm at a time. A nozzle dragging through the plate will scar the PEI coating, which is a consumable and is not covered by the warranty.

5. Loading and unloading filament

Loading

1. Choose *Filament → Load* and pick the material. The printer heats to the loading temperature for that material.
2. Cut the end of the filament at 45 degrees. A blunt or bent end will not pass the runout sensor.
3. Push the filament through the sensor and into the extruder until you feel it grip, then press Continue.
4. Wait until the colour coming out of the nozzle is the colour you loaded, then wipe the nozzle with the brass brush.

Unloading

Choose *Filament → Unload*. The printer heats, purges a few millimetres to free the tip, then retracts. Pull the filament out by hand as it retracts; do not yank it while the nozzle is cold.

6. Maintenance schedule

Interval	Task
Every print	Wipe the plate with isopropyl alcohol; brush the nozzle.
Every 50 hours	Wash the plate with soap and water. Check belt tension.
Every 200 hours	Clean and oil the linear rails with the supplied grease. Vacuum the fan grilles.
Every 400 hours	Replace the nozzle. Inspect the PTFE liner and replace it if the end is discoloured.
Yearly	Check every screw on the gantry and the bed carriage for tightness.

Belt tension is right when a plucked belt sounds like a low hum rather than a slack thud. There is no gauge and none is needed.

7. Firmware

Firmware is updated from the microSD card. Download the image that matches the mainboard revision printed beside the SD slot, copy it to the root of a card formatted FAT32, insert the card and power the printer on. The screen shows a progress bar for about ninety seconds. Do not cut the power during an update.

An image built for the wrong mainboard revision is refused with E-42.

8. Specifications

Build volume	256 × 256 × 256 mm
Nozzle	0.4 mm brass, rated to 280 °C
Bed	Spring steel with textured PEI, to 110 °C
Maximum speed	500 mm/s, 20 000 mm/s²
Connectivity	Wi-Fi, Ethernet, microSD
Power	350 W, 100–240 V
Weight	12.4 kg